

Week 1-2 Summary Report: Dataset Cleaning & Exploratory Data Analysis (EDA)

The dataset contains online e-commerce data transactions extracted from a real-world dataset. It contains 541,909 rows and 8 columns, namely, details such as the invoice number, stock code, description, quantity, invoice date, unit price, customer id and country. From those 8 columns, there is only 1 integer column, the quantity column, which makes sense as you cannot purchase 1.5 or 0.5 of a product. This is also one of the numerical features.

There are 2 float columns, which represent the other numerical features (UnitPrice and CustomerID), and the remaining are considered categorical features but the true categorical feature would be 'Country'. CustomerID can't really be considered a float as you can only have a solid number for customer ID, suggesting further future cleaning or assessment must be considered when assessing the column.

The 'InvoiceDate', 'InvoiceNo', 'StockCode' columns would need to be converted to the correct formats upon cleaning. The description is free text and does not need converting. Although CustomerID and InvoiceNo would typically serve as primary keys in a normalised relational schema (e.g., separate Customers and Invoices tables), in this flat transactional dataset they function as foreign keys, since each value repeats across multiple rows. No column in this dataset, as structured, qualifies as a true unique identifier on its own.

The dataset had missing values in Description (1,454 rows, 0.27%) and CustomerID (135,080 rows, 24.93%). Description rows were dropped since they were negligible. CustomerID rows were retained because they likely represent guest transactions. Dropping a quarter of the data would understate sales. These rows were only excluded when an analysis specifically required a customer identity.

There were 5,268 fully duplicated rows, which were removed. InvoiceDate was changed from text to a proper date format, and Description was corrected for inconsistent spacing and casing.

Quantity had 9,725 negative values. Most, 9,251, were legitimate cancellations. The remaining 474 lacked a cancellation marker, had a price of zero, and included descriptions like "DAMAGES" and "FAULTY." These indicated internal corrections and were removed.

UnitPrice contained extreme values up to £38,970, which were linked to non-product entries like "MANUAL," "AMAZON FEE," "POSTAGE," and "ADJUST BAD DEBT." These entries were removed, reducing the maximum price to £1,100.44. Rows with a price of zero were kept as genuine free items and flagged separately.

Large Quantity values over 80,000 were verified and confirmed as real wholesale orders, so they were retained. In the Country column, "EIRE" was changed to "Ireland," while "Unspecified" and "European Community" remained unchanged.

After the cleaning process, the dataset decreased from 541,909 rows and 8 columns to 532,072 rows and 9 columns.